Homework 3
of
STAT 5353 Probability and Statistics for Data Science and Bioinformatics

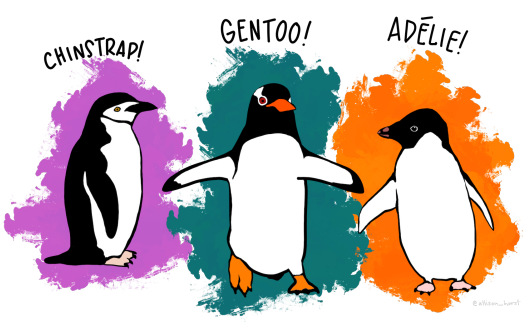
Due: 11:59 pm

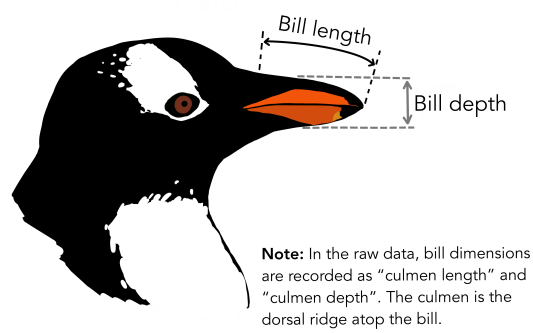
March 1 (Sunday), 2026

Problem 1 (30 points)

Work on the `penguins` dataset in the package `palmerpenguins`. This dataset contains body measurements for three penguin species observed on three islands. The description of the $p = 8$ variables among $n = 344$ observations are listed below.

- `species`: Adelie, Chinstrap, Gentoo
- `island`: Torgersen, Biscoe, Dream
- `bill length` in mm
- `bill depth` in mm
- `flipper length` in mm
- `body mass` in gram
- `sex`
- `year`





(a) Turn the variable `species` and `island` into a factor variable.

(b) Make a scatter plot between flipper length and bill length, indicating species by color.

(c) Compute the Pearson correlation coefficient between flipper length and bill length.

(d) Create three separate data sets by subsetting the `penguins` dataset in terms of the variable `species`.

(e) Compute the Pearson correlation coefficient between flipper length and bill length, separately for each Species.

(f) Recreate the plot from part (b), adding the trend lines for each species separately.

(g) Find the mean and the median of body mass of penguin in the three islands, and round the answer to two decimal places.

(h) Make density curves of the bill depth for all the three species in one plot.

(i) Make a boxplot and a violin plot of the body mass for all the three species, respectively.

(j) Make a stacked bar plot to show the relationship between `species` and `sex` variables.

Problem 2 (35 points)

Continue to work on the Heart Disease Dataset provided in Homework 2.

(a) Turn the variable `thal` into a factor variable.

(b) Turn the variable `cp` into a factor variable with four labels: "Typ", "Aty", "Non", and "Asy".

(c) Create a new factor variable `cholesterol_level` based on cholesterol levels for adults, where levels below 200 mg/dL are considered "Desirable," levels between 200 and 239 mg/dL are considered "Medium" (borderline high), and levels of 240 mg/dL and above are considered "High".

(d) Create a new factor variable `blood_level` based on resting blood pressure levels for adults, where levels below 120 mm Hg are considered "Normal" systolic, levels between 120 and 139 mmHg are considered "At Risk" (prehypertension) systolic, and levels of 140 mmHg or higher are considered "High" (blood pressure hypertension systolic)

(e) Draw a bar plot of the variable `thal` and arrange the bars in decreasing order in terms of the number of patients.

(f) Make a boxplot of cholesterol levels for female and males patients, respectively, and report their means, medians, standard deviations, and IQRs.

(g) Make a histogram of resting blood pressures for female and male patients, respectively, and describe their shapes.

Problem 3 (35 points)

Work on the `diamonds` dataset in the package `ggplot2`. The description of the $p = 10$ variables among $n = 53,940$ observations are listed below

- `price`: price in US dollars
- `carat`: weight
- `cut`: cut quality (Fair, Good, Very Good, Premium, Ideal)
- `color`: color from J (worst) to D (best)
- `clarity`: clarity from I1 (worst), SI2, SI1, VS2, VS1, VVS2, VVS1, to IF (best)
- `x`: length in mm
- `y`: width in mm
- `z`: depth in mm
- `depth`: total depth percentage
- `table`: width of top of diamond relative to widest point

(a) Turn the variables `cut`, `color`, `clarity` into factor variables

(b) Calculate the Pearson correlation coefficient between `price` and each of the variables `carat`, `depth`, `table`, `price`, `x`, `y`, and `z`.

(c) Make a bar plot of the resulting correlations in part (b) and arrange the variables in decreasing order in terms of their correlation with the `price`.

(d) Make a scatter plot where y-axis represents `price` and x-axis represents the variable with the largest Pearson correlation coefficient to `carat`. Add a trend line and interpret the meanings of intercept and slope in this context.

(e) The main colors of the diamonds are G, E and F, since they have larger numbers of samples than any others. Make density curves of `price` to compare with color in one plot and describe their shapes, respectively.

(f) Make three side-by-side box plots to compare price of diamond among the categorical variable, including `color`, `cut`, and `clarity`.

(g) Make a stacked bar plot to show the relationship between `color` and `cut`.